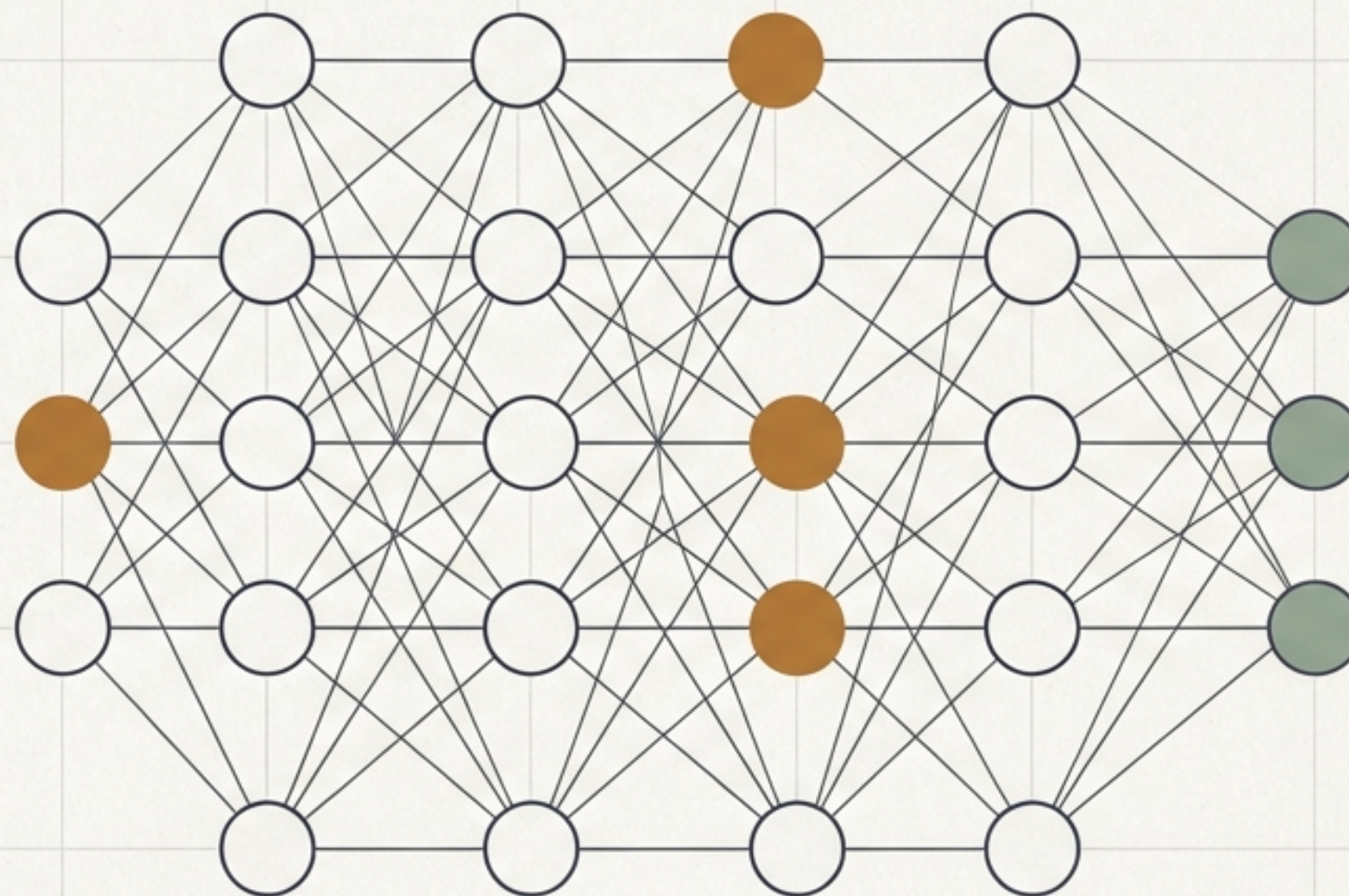


Demystifying AI: From Data to Generation.

A structural map of Machine Learning, Deep Learning, and the Generative shift.

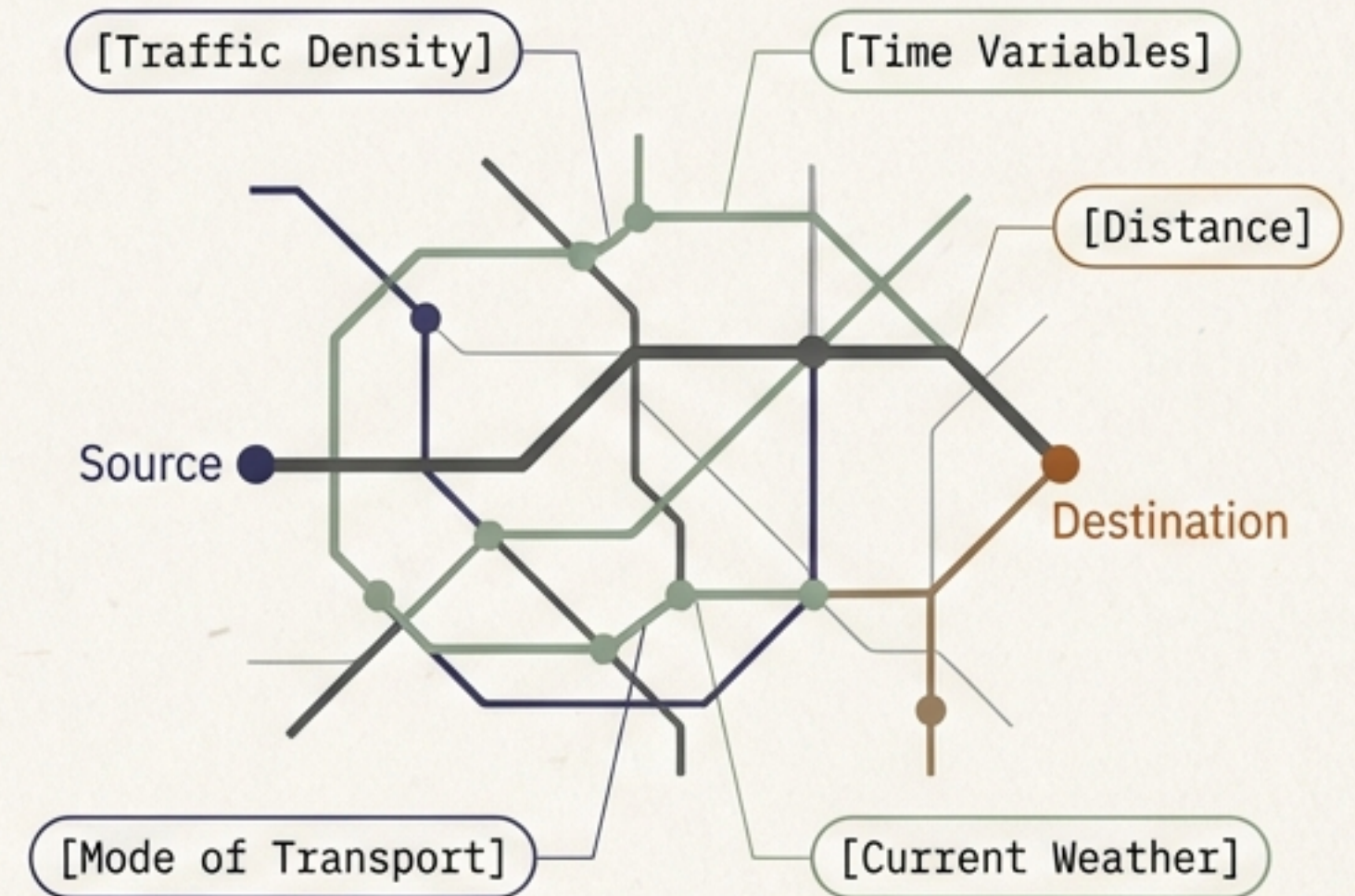


AI is machine logic mimicking human problem-solving at an incredible scale.



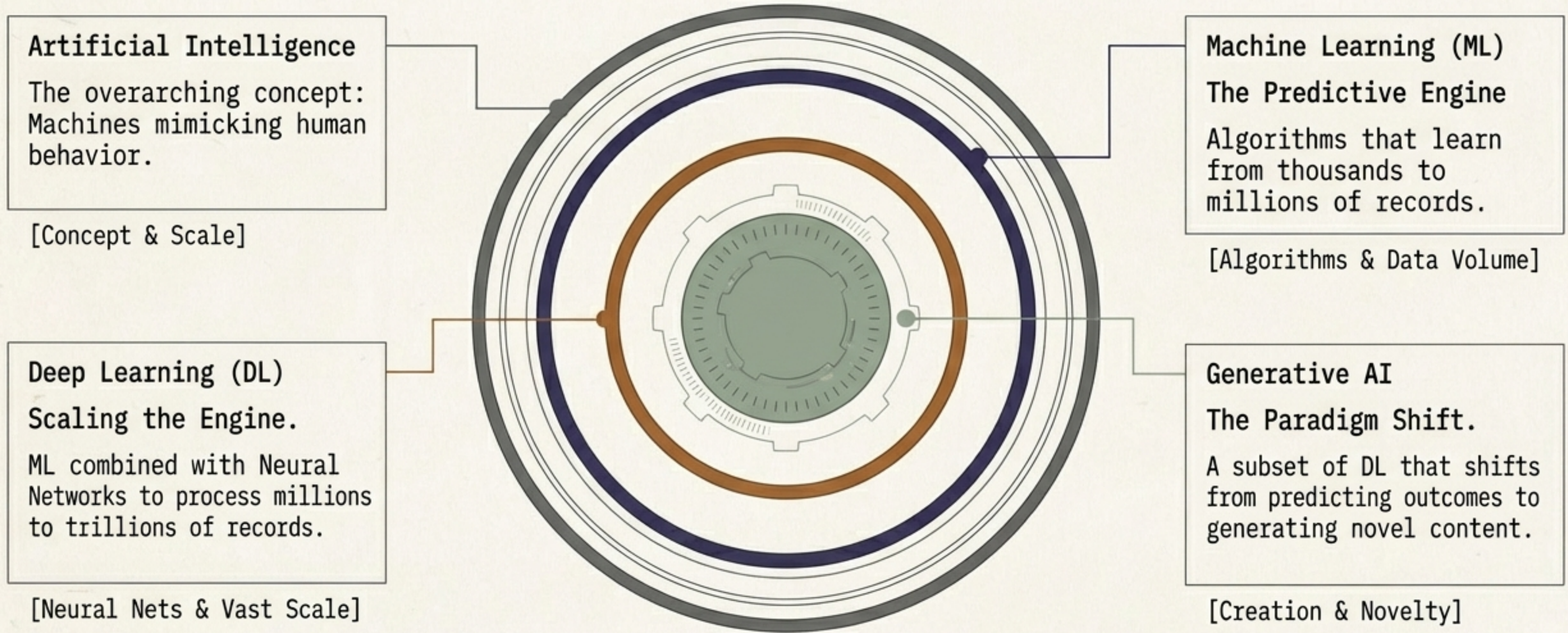
I need the shortest route to Velachery from Adyar at 9 AM in a car.

Humans rely on limited local knowledge and intuition.

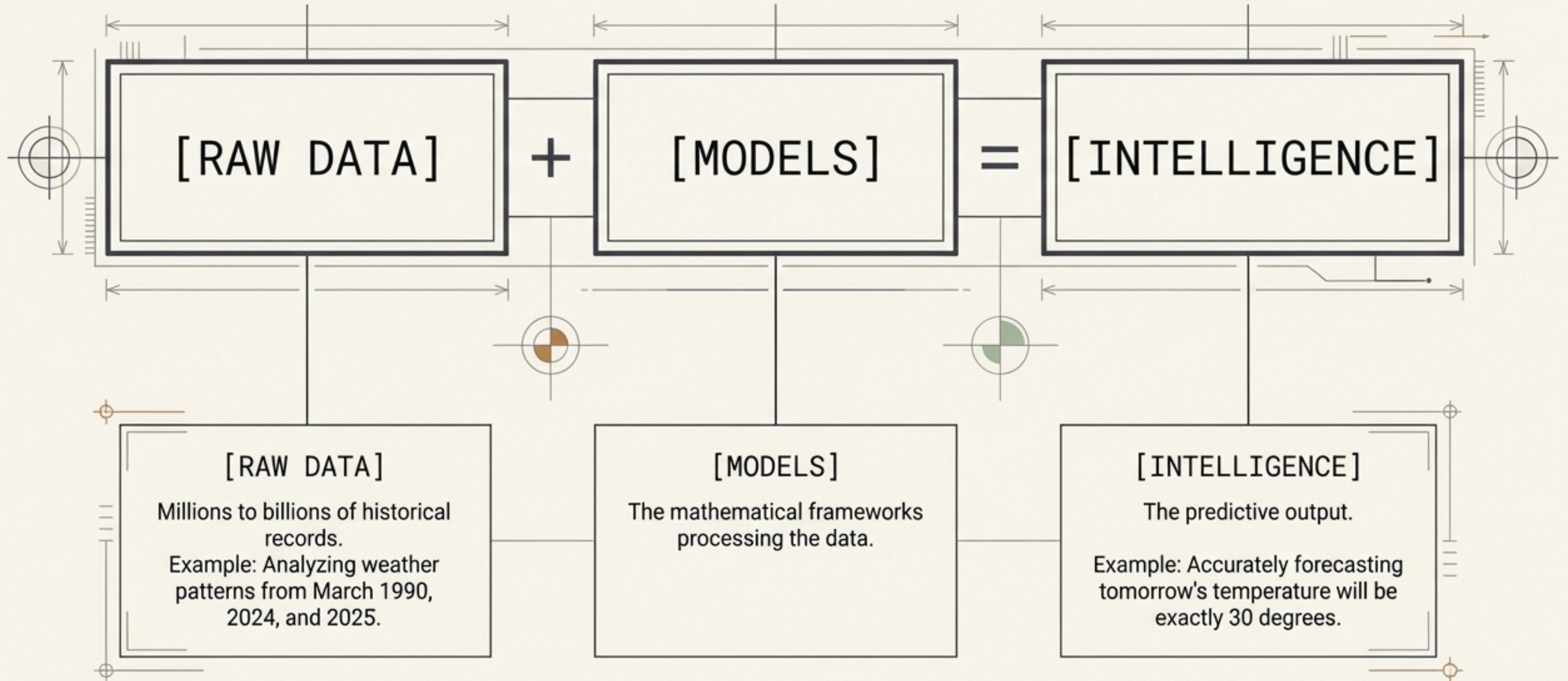


Machines achieve 90-95% accuracy in seconds by calculating millions of historical and current records.

The Nested Ecosystem of Intelligence

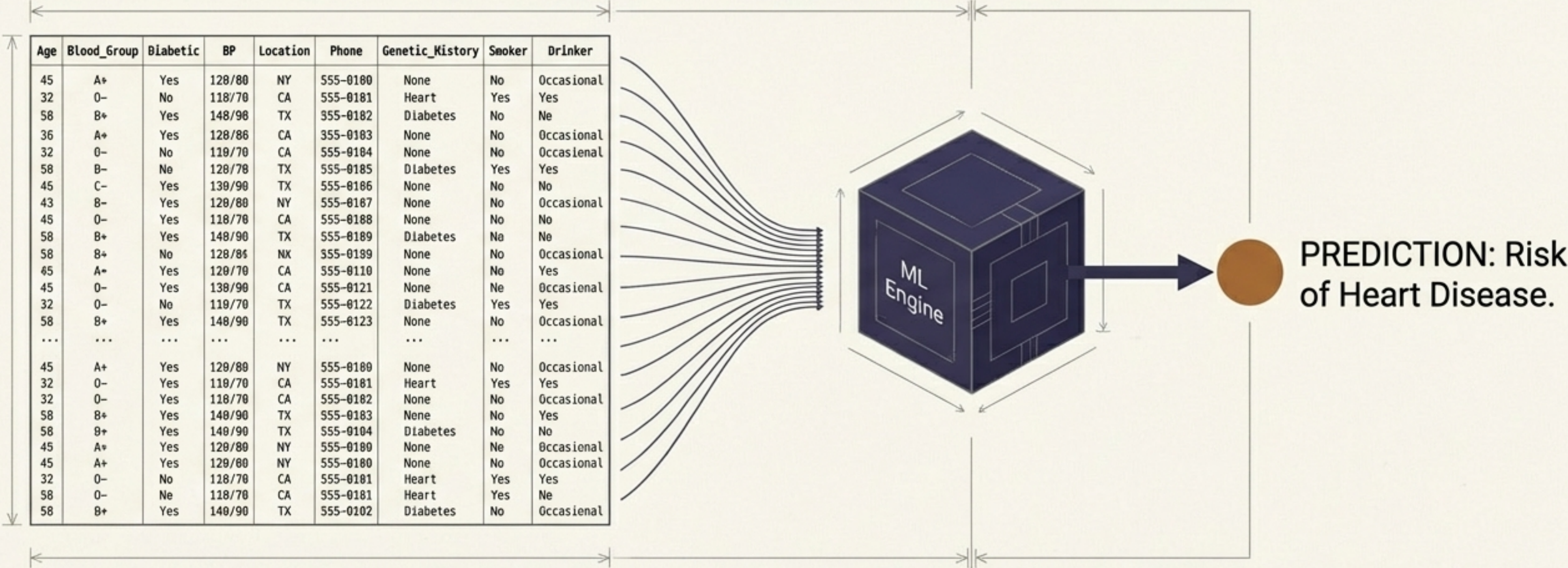


The Fundamental Engine of All AI Models

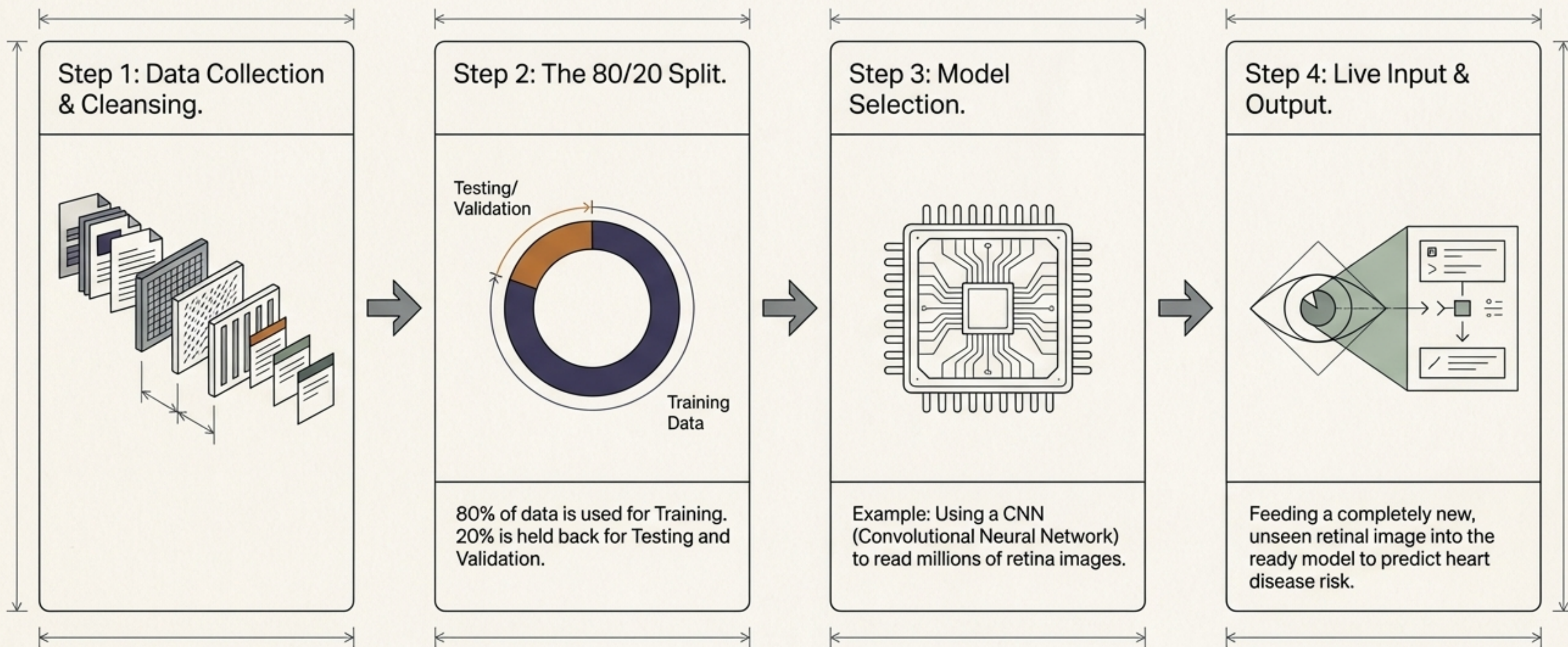


Layer 1: Machine Learning (The Predictive Engine)

ML is a subset of AI that relies on structured historical data (1,000s to millions of records) to make specific predictions.



The ML Pipeline: How models are trained and proven.

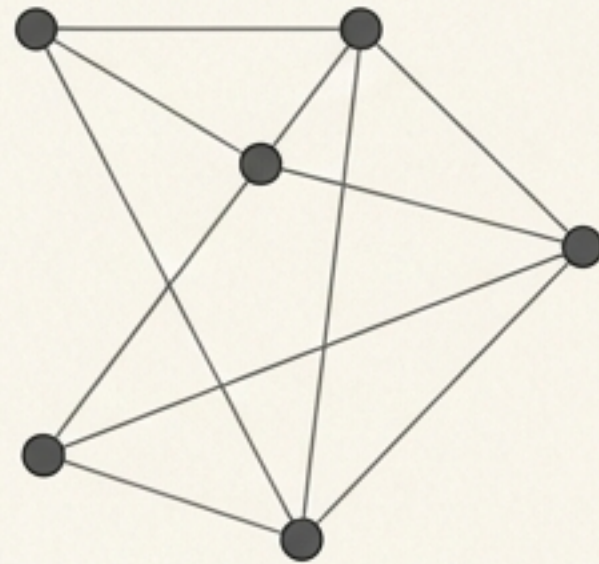


Machine Learning Mechanisms

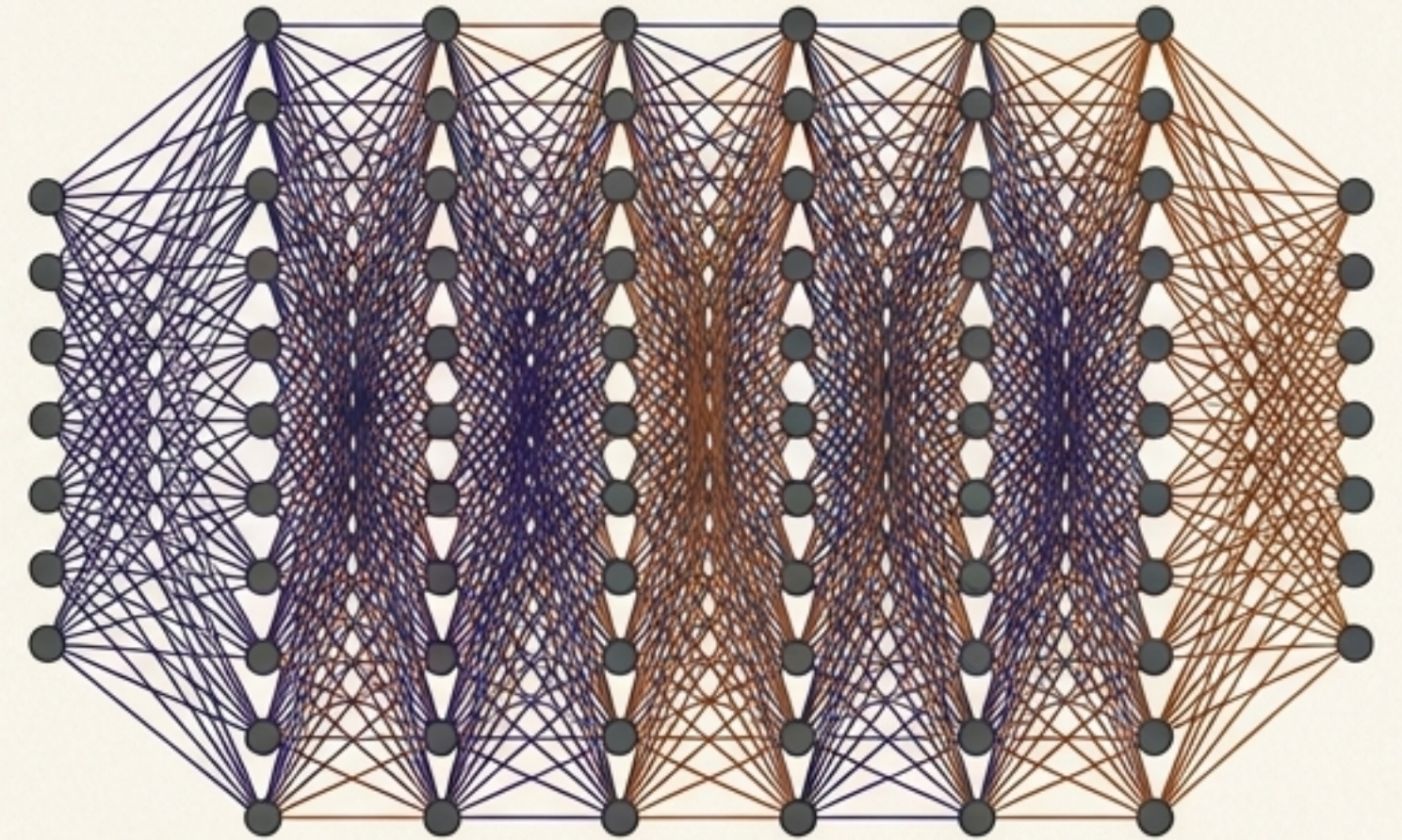
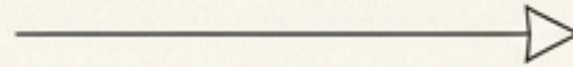
Supervised Learning	Unsupervised Learning
Data Type: Labeled (The answers are known). Example: Star ratings.	Data Type: Unlabeled.
Type A: Linear Regression. Outputs a numerical value. Example: Real Estate home purchase pricing or Defect Prediction.	Mechanism: Finding hidden patterns and grouping behaviors.
Type B: Classification. Outputs labels or categories. Example: Google predicting risk of heart attack as "Yes" or "No".	Example: Collaborative filtering based purely on user behavior, without explicit "likes" or comments.

Layer 2: Deep Learning (Scaling the Predictive Engine)

Machine Learning + Neural Networks + Vast Data = Deep Learning



Basic ML

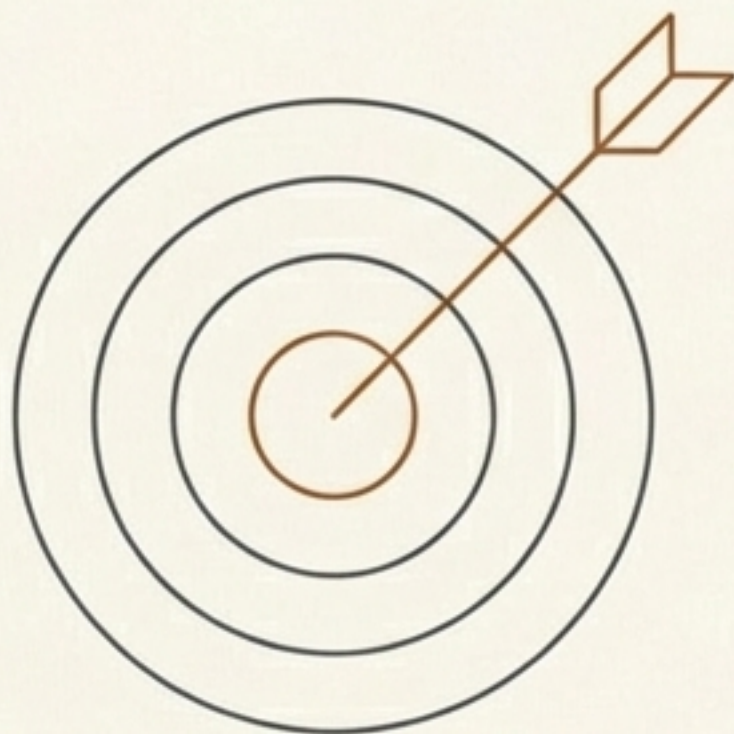


Deep Learning (Immense Scale)

Deep Learning is a predictive subset of ML built to handle immense complexity. When datasets scale from thousands of rows to millions and trillions of records, standard ML breaks down. DL utilizes layered artificial neural networks to process this vast data.

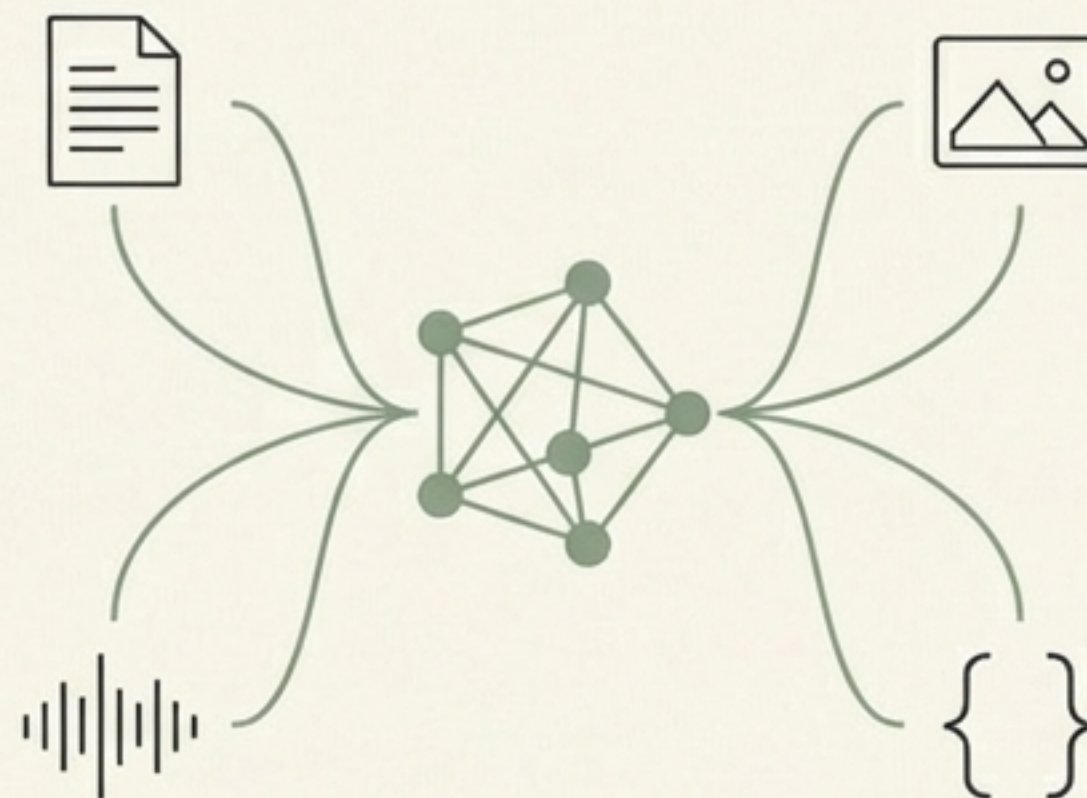
Layer 3: Generative AI (The Paradigm Shift)

Predictive AI

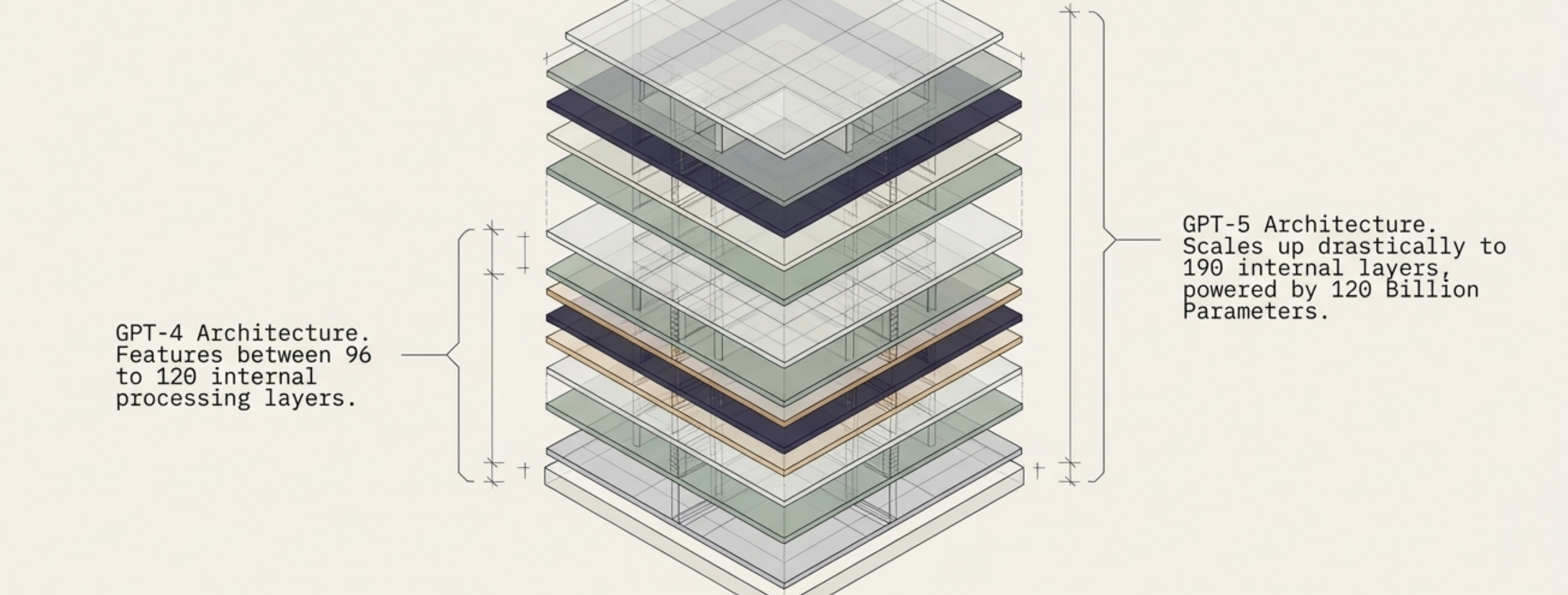


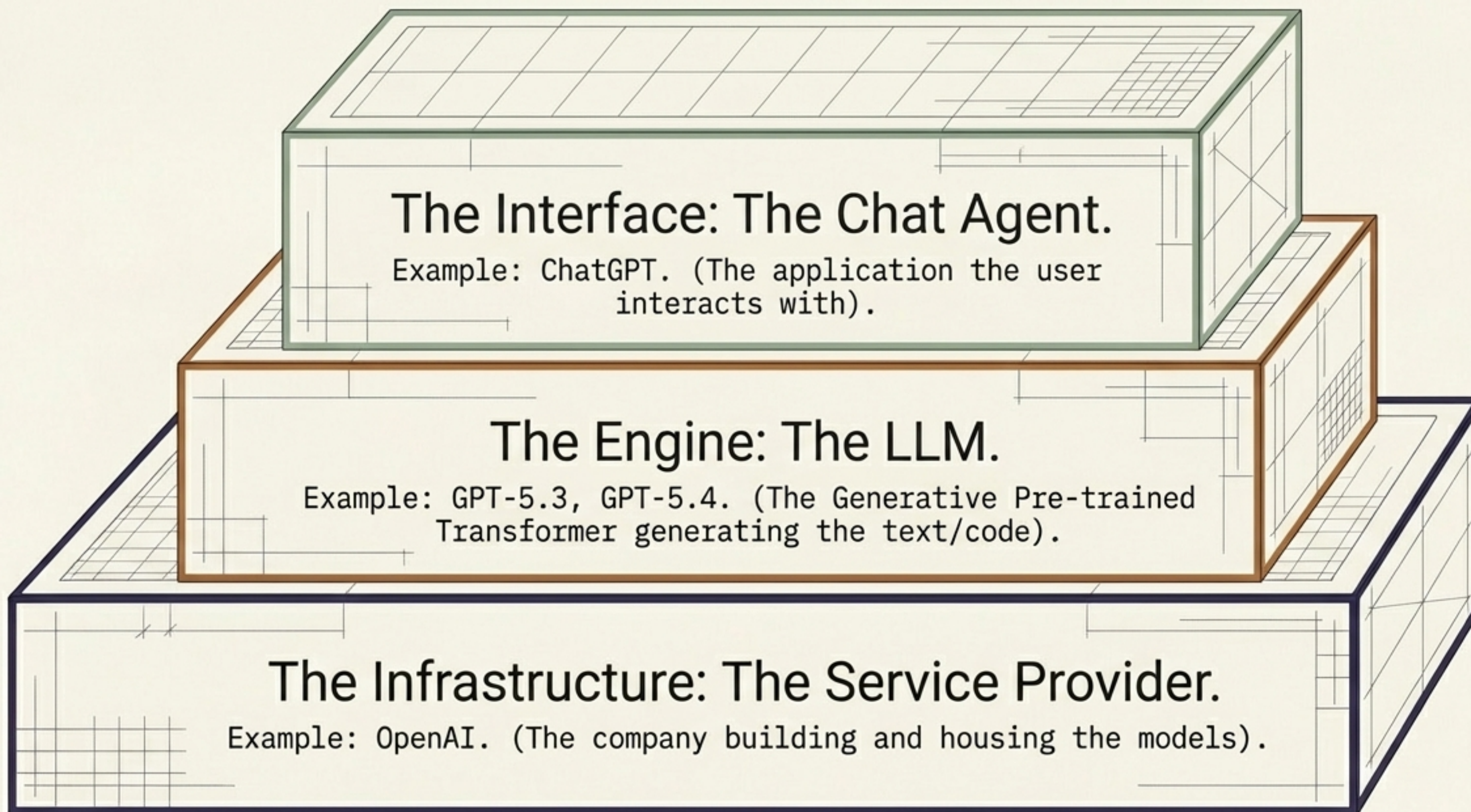
Analyzes existing data to guess a single outcome (a number, a category, or a route).

Generative AI

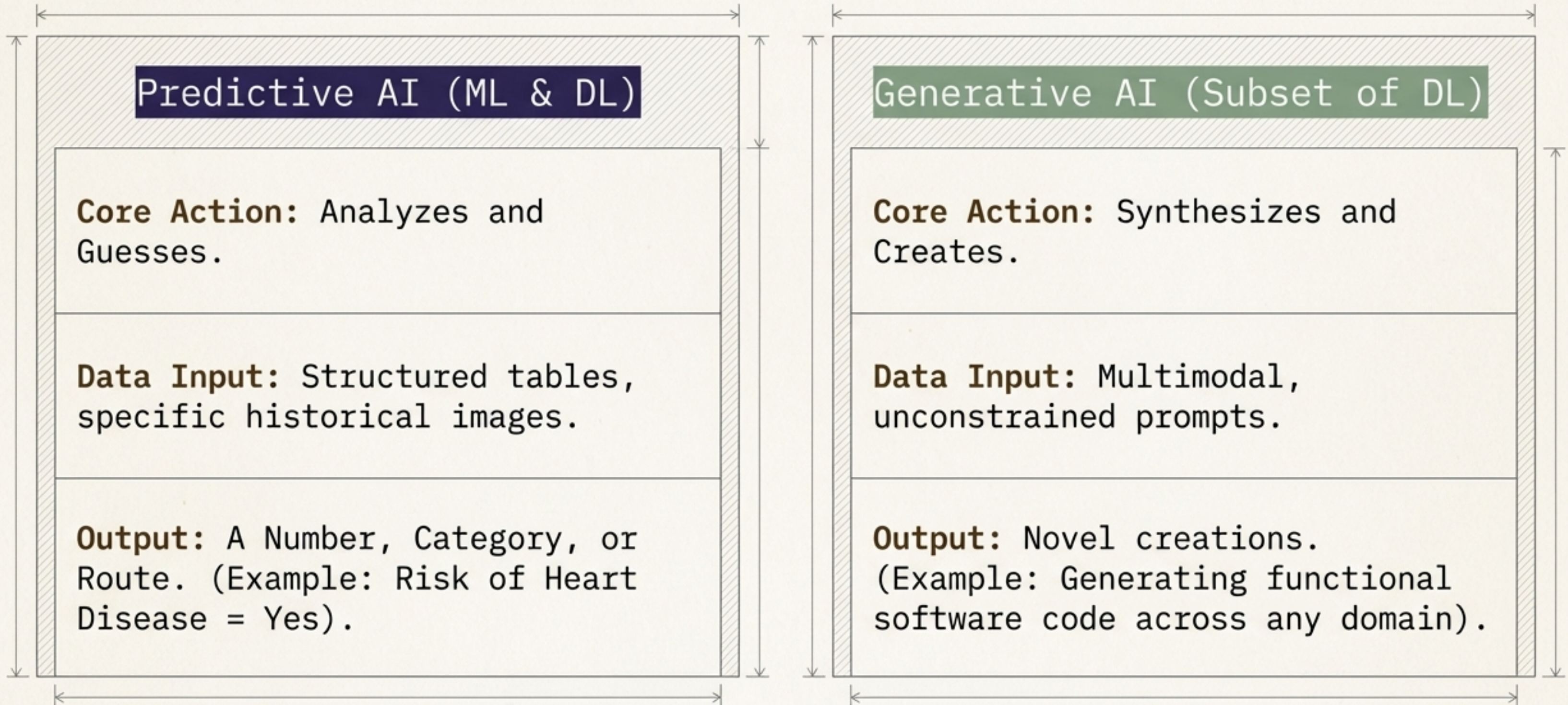


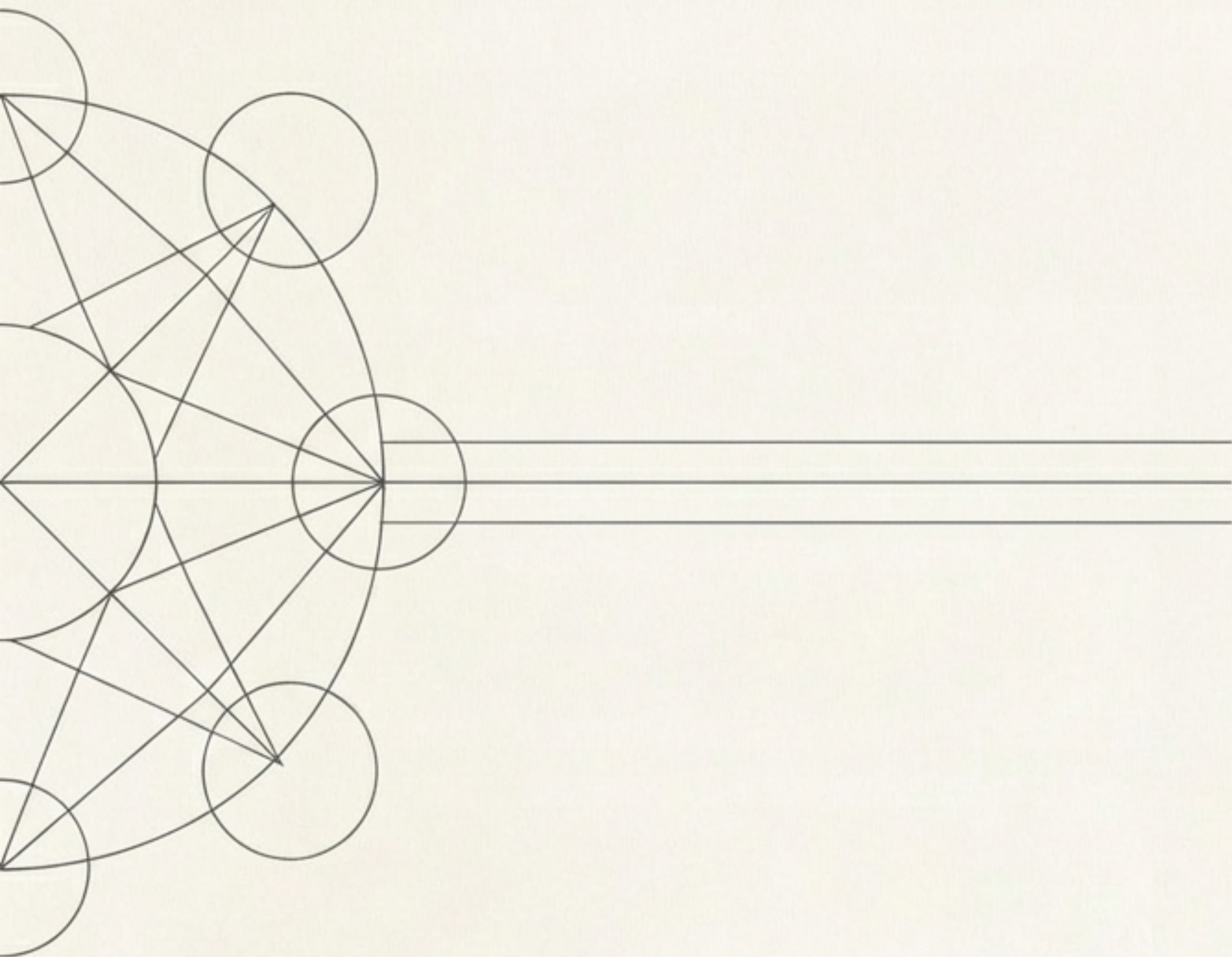
A specialized subset of Deep Learning that breaks the predictive mold. It uses vast data to generate entirely new, novel content across all domains (Image, Text, Audio, Video, and Code).





Synthesis: Predictive vs. Generative Architectures





Looking Ahead: The Engine in Detail

We have mapped the landscape from raw data to generative models. Next, we will dismantle the engine itself to explore the internal anatomy and deep working mechanics of Large Language Models.